Tutorial 5: Confidence intervals (cont.)

1. The distribution of quarterly gas bills for a particular town is thought to follow a normal distribution with mean μ_1 and variance σ_1^2. Similarly the distribution of quarterly electricity bills for the town is thought to follow a normal distribution with mean μ_2 and variance σ_2^2. A random sample of 50 gas bills and 60 electricity bills is considered and it is found that the mean amounts are £160 and £180 respectively. Suppose that it is known that the standard deviation in all gas bills is 30 and the standard deviation in electricity bills is 40. Find a 95% confidence interval for the difference in population means between gas and electricity.

2. Let $\underline{X}$ be a random sample of size n from the normal distribution $\mathrm{N}(\mu, \sigma^2)$.

   (a) Using $\frac{(n-1)S^2}{\sigma^2}$ as a pivotal quantity, develop an approximate 95% CI's for σ^2 by using the *large sample normal approximation* for χ^2_{n-1} (given in lectures; see Remark 4.3 in the lecture notes).

   (b) Again, using $\frac{(n-1)S^2}{\sigma^2}$ as a pivotal quantity, develop an approximate 95% CI for σ^2 by using the approximation for the χ^2_{n-1} distribution which is given in the statistical tables (NCST, page 40).

   (c) Suppose that $n = 101$ and $S^2 = 1$. Evaluate the 95% CI for σ^2 obtained above and compare your answers with the 'exact' 95% CI for σ^2 which you can obtain from R using, e.g.,qchisq(0.025,100) .

3. In order to assess the effect of washing on the extensibility of yarn, twenty lengths are taken. Ten are selected at random to serve as controls and the remaining ten are put through a washing cycle. All are measured for breaking extension given as a percentage of the length. The resulting measurements are:

   Control: 15 15 10 18 17 13 9 12 12 13

   Washed: 13 16 8 12 14 11 13 14 9 11

   (a) Determine a 95% confidence interval for the ratio of the two corresponding population variances σ_c^2/σ_w^2.

   (b) Determine a 95% confidence interval for the mean decrease in extensibility due to washing.

   (c) Discuss the validity of any assumptions required in your two analyses.

4. A laboratory experiment is performed to compare two methods of evaluating the coefficient of thermal expansion of a polymer. Samples of eighteen varying grades of a polymer are used, nine with method A and nine with method B, yielding the following results:

Method A	Method B
6.8	7.2
8.4	8.3
7.9	8.3
6.5	7.5
9.5	9.5
7.0	7.7
9.1	9.4
7.5	7.8
9.8	10.3

   (a) Determine a 95% confidence interval for the difference between the mean for method B and method A.

   (b) It is subsequently discovered that only nine samples were used, and both methods were applied to each sample, i.e. the data are paired with each row of the table coming from one sample. By considering the nine differences, one for each pair, as a single sample from an appropriate population, develop a 95% confidence interval for the difference between the mean for method B and the mean for method A.

   Comment on the two analyses.

5. A survey is undertaken to investigate the proportion p in the adult population who support a certain government policy. Determine a 95% confidence interval for p when

   (a) a sample of 10 adults contains 3 who support the policy;

   (b) a sample of 100 adults contains 30 who support the policy.

   Comment on your results.

Exam Style Question

Consider two normally distributed populations X and Y with means μ_X and μ_Y and variances σ_X^2 and σ_Y^2 respectively. Also consider two independent random samples, one from each population, given by $X_1, \ldots, X_{n_X}$ and $Y_1, \ldots, Y_{n_Y}$ respectively.

(a) Show that $\bar{X} - \bar{Y}$ is an unbiased estimator for $\mu_X - \mu_Y$.

[2 marks]

(b) Explain why, when the variances are known, the function

$$h(\underline{X}, \underline{Y}, \mu_X - \mu_Y) = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}}$$

is a pivotal quantity for the difference $\mu_X - \mu_Y$.

Note: You may quote any standard results.

[4 marks]

(c) Suppose now that $\bar{x} = 65, \bar{y} = 70, n_X = 12, n_Y = 10$. But this time the population variances are **not** know but we do know that $s_X = 4.5$ and $s_Y = 3.1$. Calculate an equal-tailed 95% confidence interval for the difference between the population means specifying the pivotal quantity used in this case.

[4 marks]

[Total 10 marks]